

SCUOL, Switzerland

WMO: 067980

Lat: 46.7933N

Lon: 10.2833E

Elev: 1295

StdP: 86.70

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.9	(c) -13.0	(d) -21.3	(e) 0.7	(f) -13.6	(g) -18.4	(h) 0.9	(i) -11.6	(j) 6.9	(k) -5.3	(l) 5.5	(m) -5.6	(n) 2.0	(o) 260	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	28.1	15.3	26.3	14.8	24.5	14.2	16.7	24.6	15.8	23.3	15.0	22.1	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	11.8	17.9	13.3	11.2	17.2	12.6	10.6	16.6	51.8	24.8	49.0	23.4	46.6	22.0	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.0	(b) 5.0	(c) 4.3	(e) -17.8	(f) 31.0	(g) 2.6	(h) 1.4	(i) -19.7	(j) 32.0	(k) -21.2	(l) 32.8	(m) -22.7	(n) 33.5	(o) -24.5	(p) 34.5		
			DB													
			WB													
				-18.5	18.4	2.5	0.7	-20.3	18.9	-21.8	19.3	-23.2	19.7	-25.0	20.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-4.2	-2.3	2.6	7.1	10.7	15.0	16.4	15.9	11.8	7.8	1.8	-3.1
	DBStd	8.16	3.92	4.44	4.15	3.52	3.48	3.66	3.24	3.34	3.46	3.77	4.13	4.11
	HDD10.0	1908	439	345	229	97	34	5	1	1	22	86	245	405
	HDD18.3	4291	697	578	487	337	238	108	77	88	196	326	494	663
	CDD10.0	699	0	0	0	11	54	156	198	186	76	18	0	0
	CDD18.3	39	0	0	0	0	1	9	16	14	0	0	0	0
	CDH23.3	629	0	0	0	0	13	173	233	191	19	1	0	0
	CDH26.7	122	0	0	0	0	1	33	47	40	1	0	0	0
(14) Wind	WSAvg	1.5	1.3	1.4	1.6	1.9	1.9	1.8	1.7	1.5	1.4	1.2	1.1	1.3
(15) Precipitation	PrecAvg	767	41	37	40	40	64	92	107	113	69	64	59	41
	PrecMax	986	103	174	104	84	127	150	179	201	140	152	194	142
	PrecMin	587	4	9	8	8	24	46	70	44	25	4	5	5
	PrecStd	98	27	36	24	17	24	28	26	35	31	35	40	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	10.5	17.0	21.1	25.1	29.8	30.3	30.1	25.8	20.8	13.7	6.5
		MCWB	3.1	3.8	6.5	9.8	12.9	15.4	16.3	15.5	14.1	11.3	7.8	3.0
	2%	DB	3.7	7.4	14.5	18.9	22.5	27.5	28.0	27.6	23.1	18.1	10.9	4.5
		MCWB	1.1	2.0	5.4	8.5	11.9	14.9	15.4	15.6	12.8	10.5	6.5	1.7
	5%	DB	2.1	5.2	12.2	16.6	20.3	25.4	26.1	25.5	20.9	16.0	8.9	3.1
		MCWB	0.3	0.9	4.5	7.5	10.8	14.3	14.7	15.0	12.4	9.7	5.4	0.9
	10%	DB	0.8	3.2	9.5	14.5	18.0	23.1	24.0	23.1	18.5	13.8	7.0	1.8
		MCWB	-0.6	0.1	3.4	6.6	9.8	13.4	14.3	14.4	11.9	8.9	4.5	0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	4.4	7.4	10.3	13.9	17.2	17.7	17.8	15.4	12.6	8.7	4.2
		MCDB	5.2	8.9	14.7	19.4	22.7	26.1	26.0	25.2	22.3	18.7	12.0	5.6
	2%	WB	1.7	2.6	6.0	9.1	12.6	15.9	16.6	16.7	14.2	11.5	7.1	2.6
		MCDB	3.1	6.1	12.8	17.5	20.4	24.3	24.6	24.3	20.4	16.2	9.7	3.7
	5%	WB	0.6	1.4	4.9	8.1	11.6	14.9	15.7	15.7	13.3	10.4	5.9	1.4
		MCDB	1.8	4.3	11.0	15.2	18.5	22.9	23.2	22.8	18.7	14.5	8.1	2.4
	10%	WB	-0.3	0.4	3.8	7.1	10.5	14.0	15.0	14.9	12.3	9.5	4.8	0.4
		MCDB	0.8	2.8	8.8	13.1	16.6	21.5	21.9	21.3	17.1	13.0	6.7	1.5
(35) Mean Daily Temperature Range	MDBR		7.0	9.3	11.4	12.0	11.7	12.8	12.8	12.3	11.5	10.7	7.7	6.3
	5% DB	MCDBR	7.4	10.9	15.5	15.9	15.6	16.4	16.5	16.4	15.6	13.6	10.1	7.0
		MCWBR	5.7	7.1	8.1	7.5	6.9	6.5	6.3	6.4	7.1	7.4	6.4	5.0
	5% WB	MCDBR	6.4	9.3	13.7	14.4	13.8	14.7	14.3	14.0	13.0	11.6	8.9	5.8
		MCWBR	5.3	6.3	7.3	7.0	6.3	6.2	6.1	6.1	6.4	6.7	6.0	4.6
(40) Clear-Sky Solar Irradiance	taub		0.243	0.268	0.313	0.364	0.379	0.386	0.383	0.376	0.341	0.312	0.275	0.243
	taud		2.335	2.266	2.211	2.201	2.264	2.296	2.306	2.345	2.427	2.480	2.466	2.381
	Ebn at Noon		868	907	908	889	885	878	875	868	872	850	824	828
	Edh at Noon		72	96	119	134	130	127	124	115	96	78	64	61
(44) All-Sky Solar Radiation	RadAvg		1.54	2.48	3.68	4.45	4.87	5.57	5.58	4.88	3.75	2.52	1.52	1.27
	RadStd		0.17	0.29	0.45	0.63	0.52	0.47	0.45	0.46	0.39	0.31	0.24	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	N/A
(47) Regional (107 neighbors)	+0.34	N/A	N/A	+0.64	+0.42	+0.36	N/A	N/A	+59	+16

Nomenclature: See separate page